

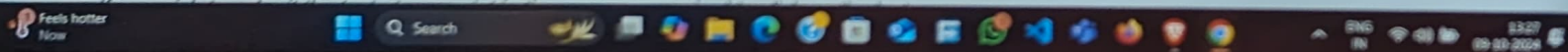
Welcome to Qualcomm Assessment

Test Duration: 90 Minutes

Important Instructions:

"Every correct answer carries 1 mark and wrong answer carries negative 0.25 marks".

- Entire duration of the Assessment will be Video & Audio Recorded and Proctored. Please be available in front of the camera throughout the test duration.
- Please ensure that you are taking the test where your face has enough lighting.
- Kindly refrain from using the REFRESH button on the screen at any time during the test.
- If you are trying to navigate from the test window or from the camera, the system will give you an alert.
- Do not use ALT+TAB during the test or navigate away from the test page. Doing so 3 times will automatically log you out from the test.



TIME LEFT: 00:30:00

QUESTION 1 GROUP Aptitude SECTION Puzzles Mark(s) 1

'R' is the only sister of 'P's husband, who is the only son of 'L's grandmother. 'Q' is the son of 'W', who is the brother-in-law of 'S' and only son-in-law of 'R's mother. 'L' is not a child of 'R'. How is 'R' related to 'Q's daughter'?

☒ Mother☐ Aunt☐ Grandmother☐ Cousin

Previous Question

Next Question

TIME LEFT: 00:29:34

1 ☆ QUESTION: 1 GROUP: Aptitude SECTION: Puzzles Mark(s): 1

2 'R' is the only sister of P's husband, who is the only son of L's grandmother. 'Q' is the son of 'W', who is the brother-in-law of 'S' and only
3 son-in-law of R's mother. 'L' is not a child of 'R'. How is 'R' related to Q's daughter?

☒ Mother☐ Aunt☐ Grandmother☐ Cousin

Previous Question

Next Question

TIME LEFT: 00:26:31

QUESTION 2 GROUP: Aptitude SECTION: Puzzles Mark(s): 1

In a certain code, ALBUM is written as WODCN. How is CRAFT written in that code?

☐ HVCET☐ HVECT☐ HCEVT☐ HEVCT

✱

✓

ALBUM
+2
WODCN

Previous Question

Next Question

TIME LEFT: 00:24:24

QUESTION 3 GROUP: Aptitude SECTION: Problem Solving Mark(s): 1

A can do a piece of work in 2 hours, B can do thrice the work in 8 hours and C can do the same work as A in 8 hours. If all of them work together, how long it would take them to complete the work:

☒ 1 hour☐ 3 hours☐ 4 hours☐ 2 hours

Previous Question

Next Question

TIME LEFT: 00:24:22

QUESTION 3 GROUP: Aptitude SECTION: Problem Solving Mark(s): 1

A can do a piece of work in 2 hours, B can do thrice the work in 8 hours and C can do the same work as A in 8 hours. If all of them work together, how long it would take them to complete the work:

☒ 1 hour☐ 3 hours☐ 4 hours☐ 2 hours

$$\frac{1}{8} + \frac{3}{8} + \frac{1}{8}$$

Previous Question

Next Question

TIME LEFT: 00:22:15

QUESTION: 4 GROUP: Aptitude SECTION: Problem Solving Mark(s): 1

'A', 'B' and 'C' can complete a job in 20 days and 35 days respectively and 'A', 'B' and 'C' can complete the same job in $8\frac{3}{4}$ days working together. 'A' is twice as efficient as 'D'. They start doing the job such that only one person works in one day starting with 'D' followed by 'B', 'A' and 'C'. If 'A' was paid Rs. 210 per day, then how much was 'D' paid for the job?

☐ Rs. 735☐ Rs. 720☐ Rs. 840☐ Rs. 630

LOOK INTO NOTES

Previous Question

Next Question

A cyclist moving on a circular track of radius 100 metres completes one revolution in 2 minutes. What is the approximate speed of the cyclist?

☐ 316 m/min

☐ 319 m/min

☒ 314 m/min

☐ 214 m/min

$$2 \times \pi \times 100 = 2 \text{ min}$$

1 min

$$= \frac{2\pi \times 100}{2}$$

Previous Question

Next Question

TIME LEFT: 00:19:00

QUESTION: 6 GROUP: Aptitude SECTION: Logical Reasoning Mark(s): 1

Direction: Study the information given below and answer the questions that follow.

Six friends - A, B, C, D, E, and F participated in a quiz competition and each scored different marks. The one who scored highest is ranked 1 and so on. The following information is known about their scores:

- A scored higher than B but lower than C.

- D scored higher than E but lower than F.

- F scored the highest.

- E scored higher than A and C scored less than E.

Question: Who scored the lowest?

☐ Cannot be determined

B

Previous Question

Next Question

TIME LEFT: 00:16:45

2

QUESTION 7 GROUP: Aptitude SECTION: Logical Reasoning Mark(s): 1

3

Direction: Study the information given below and answer the questions that follow.

4

Six friends - A, B, C, D, E, and F participated in a quiz competition and each scored different marks. The one who scored highest is ranked 1 and so on. The following information is known about their scores:

5

6

- A scored higher than B but lower than C.

- D scored higher than E but lower than F.

7 ☆

- F scored the highest.

8

- E scored higher than A and C scored less than E.

Question: Who scored the third highest?

9

☐ D

10

☐ A

E

TIME LEFT: 00:15:21

A scored higher than B but lower than C.

D scored higher than E but lower than F.

F scored the highest.

E scored higher than A and C scored less than E.

Question: What will be the difference between the ranks of A and F?

1

4

3

2



Previous Question

Next Question

TIME LEFT: 00:14:27

QUESTION 11 GROUP: Aptitude SECTION: Problem Solving Mark(s): 1

An iron pipe has a thickness of 2 cm, length of 28 cm and external diameter of 6 cm. If it is given that the weight of iron is 12 g/cm^3 then find the weight of the iron pipe. (1 cm^3 of iron weights 12 gm)

☐ 4350 gm☐ 5280 gm☐ 4800 gm☐ 3480 gm[Previous Question](#)[Next Question](#)

TIME LEFT: 00:12:08

QUESTION: 12 GROUP: Aptitude SECTION: Problem Solving Mark(s): 1

Father is aged three times more than his son Ronit. After 8 years, he would be two and a half times of Ronit's age. After further 8 years, how many times would he be of Ronit's age?

☐ $2\frac{1}{2}$ times☐ 2 times☐ $2\frac{3}{4}$ times☐ 3 times

more

$$F = 4R$$

Previous Question

Next Question

A school has four sections A, B, C and D of Class VII students. The results of the first term exam and final exam have been shown in the table given below.

Result	Section A	Section B	Section C	Section D
No. of students passed in both the exams	76	68	66	72
No. of students failed in both the exams	23	28	25	21
No. of students passed in first term exam but failed in final exam	11	9	13	7
No. of students failed in first term exam but	15	12	8	10

Previous Question

Next Question

TIME LEFT: 00:09:40

No. of students failed in
first term exam but
passed in final exam

15

12

8

10

Question: Find the total number of students in class VII in the school.

☐ 464 students☐ 467 students☐ 469 students☐ 461 students

ALL SUM

Previous Question

Next Question

TIME LEFT: 00:06:52

QUESTION 18 GROUP: Aptitude SECTION: Problem Solving Mark(s): 1

If a die, was thrown twice, what is the probability that at least one of the two throws comes up with the number 4?

☐ 15/32☐ 5/6☒ 11/36☐ 5/18

Previous Question

Next Question

TIME LEFT: 00:04:51

QUESTION 19
GROUP: Aptitude

SECTION: Problem Solving

Mark(s): 1

In how many ways 20 students can be seated in a round table, if four particular students always get seated together?

☐ 17! 3!☐ 18! 2!☐ 15! 5!☒ 16! 4!

Previous Question

Next Question

TIME LEFT: 00:04:00

QUESTION: 20 GROUP: Aptitude SECTION: Problem Solving Mark(s): 1

If the 11th and the 8th terms of an arithmetic progression are in the ratio 2 : 5, which of the following statements is definitely false?

☐ The sum of no two consecutive terms is zero

☐ At least one of the terms is zero

☐ The sum of no three consecutive terms is zero

☐ None of the above

?

TIME LEFT: 00:02:30

QUESTION 20 GROUP: Aptitude SECTION: Problem Solving Mark(s): 1

If the 11th and the 8th terms of an arithmetic progression are in the ratio 2 : 5, which of the following statements is definitely false?

☐ The sum of no two consecutive terms is zero

☐ At least one of the terms is zero

☐ The sum of no three consecutive terms is zero

☐ None of the above

[Previous Question](#)[Next Question](#)

21

30 13 8 17 40.

The given algorithm is:

22

i) Sort the given elements in ascending order

23

ii) Push all the elements into the stack of size 5 and initially the stack is empty

24

iii) Perform the following operations

```
for (i=0; i<5; i++)
```

25

```
{  
    pop();
```

26

```
}  
iv) For all the popped elements, perform the following operations
```

27

Add 2 to even numbers

28

Add 3 to odd numbers

29

v) Insert all elements into the queue of size 5 and initially, the queue is empty

vi) Perform the following operations

```
for (i=0; i<5; i++)
```

[Previous Question](#)[Next Question](#)

TIME LEFT: 00:28:34

21 v) Insert all elements into the queue of size 5 and initially, the queue is empty

vi) Perform the following operations

22 ☆ for($i=0; i<5; i++$)

23 {

24 dequeue();

}

25 Identify the dequeued elements from the given options.

26 ☐ 10 16 20 32 42 ✓

27 ☐ 42 32 20 16 10

28 ☐ None of the options given

29 ☐ 8 13 17 30 40

Previous Question

Next Question

TIME LEFT: 00:27:38

QUESTION: 23 GROUP: Programming Basics SECTION: C Programming Mark(s): 1

Which of the code given in options can be placed at //blank to execute the below given code successfully?

int *P, *Q;

//blank

printf("%d ", *P++);



☐ P = &Q = (int *) malloc(sizeof(int));

☐ *P = &Q = (int *) malloc(sizeof(int));

☐ P = Q = (int *) malloc(sizeof(int));

☒ P = &Q(int *) malloc(sizeof(int));



Previous Question

Next Question

TIME LEFT: 00:25:27

21 QUESTION 24 GROUP: Programming Basics SECTION: Operating System Fundamentals

Mark(s): 1

22 Which of the below given statement(s) is/are correct for best fit?

23 I) The smallest hole which is big enough is allocated to program.

24 II) The largest hole which is big enough is allocated to program.

25 ☐ Both (I) and (II)26 ☒ Only (I)27 ☐ Only (II)28 ☐ Neither (I) nor (II)

29

Previous Question

Next Question

21 QUESTION: 25 GROUP: Programming Basics SECTION: Data Structures Basics Mark(s): 1

22 Which of the below given statement(s) is/are the advantage(s) of trees?

- 23
- I) It reflects the structural relationships in the data.
 - II) It is used to represent hierarchies.
 - III) It provides an efficient insertion and searching of data.

24

25

☒ All (I), (II) and (III)

26

☐ Only (I) and (II)

27

☐ Only (I)

28

☐ Only (II) and (III)

Previous Question

Next Question

TIME LEFT: 00:22:58

QUESTION: 27 GROUP: Programming Basics SECTION: Data Structures Basics Mark(s): 1

Which of the below given statement(s) is/are correct?

- I) If a queue is represented using singly linked list both insertion and deletion operations will take $O(1)$ time. ✓
- II) If a queue is represented using doubly linked list both insertion and deletion operations will take $O(n)$ time. ✗

☐ Only (II)☐ Both (I) and (II)☒ Only (I) ✓☐ Neither (I) nor (II)

Previous Question

Next Question

TIME LEFT: 00:21:18

23

24

```
int p = 10, q = 10, r = 10;
```

```
p << 2 >> 1;
```

25

```
r <<= 2 >> 1;
```

26

```
printf("%d %d %d", p, q<<2>>1, r);
```

```
return 0;
```

27

28

☐ 10 10 20

29

☒ 10 20 20 ✓

30

☐ 10 20 10

31

☐ 10 10 10

Previous Question

Next Question

TIME LEFT: 00:19:38

QUESTION: 29 GROUP: Programming Basics SECTION: Computer Architecture Mark(s): 1

In which of the below given dependency(ies) a hardware mechanism called renaming is used to minimize stalls in the pipeline?

I) Structural Dependency

II) Control Dependency

III) Data Dependency

7.

☒ Only (III)

☐ Only (I)

☐ Only (II) and (III)

☐ All (I), (II) and (III)

Previous Question

Next Question

26 ★

```
#include <stdio.h>
```

27

```
int main()
```

28 ★

```
{  
    void *ptr;
```

//line 4

29 ★

```
    int arr = 9;
```

//line 5

```
    ptr = arr;
```

//line 6

30

```
    printf("%d", *ptr);
```

//line 7

31 ☆

Which of the statement given in the option is the correct one to be replaced in the program so that it will be error free?

TIME LEFT: 00:17:38

26 ★

```
void *ptr; //line 4
```

```
int m2 = 9; //line 5
```

```
ptr = &m2; //line 6
```

```
printf("%d", *ptr); //line 7
```

void* can point to any type, but to dereference it, you must cast it to a specific type, like `int*`. `*(int*)ptr` first casts `ptr` to `int*`, and then dereferences it to access the integer value it points to.

Which of the statement given in the option is the correct one to be replaced in the program so that it will be error free?

27 What would be the output when the value assigned to the variable "d" is 10?

28 ★
#include <stdio.h>

29 ★
int main()

30
{

char *point = (char*)&d;

31 printf("%d ", ++*point++);

*point += 1;

32 ☆ printf("%d ", ++*point--);

return 0;

33
34 ☐ 12 14

35 ☐ 11 2

Previous Question

Next Question

27

28 ★

29 ★

30

31

32 ☆

33

34

35

```
char *point = (char*)0;  
printf("%d ", ++*point++);  
*point += 1;  
printf("%d ", ++*point--);  
return 0;  
}
```

☐ 12 14☐ 11 2☐ 12 13☐ 11 1

Previous Question

Next Question

28 ★

{

29 ★

int *j;

void fun(int **);

30

fun(&j);

31

return 0;

}

32 ★

void fun(int **k)

33 ☆

{

34

int a = 100 * 1000 + 100 / 5;

35

printf("%d", *k = a);

}

36



100020



```
void fun(int **k)
```

```
{
```

```
    int a = 100 * 1000 + 100 / 5;
```

```
    printf("%d", *k = a);
```

```
}
```

100020

TIME LEFT: 00:09:04

30 QUESTION: 35 GROUP: Programming Basics SECTION: Data Structures Basics Mark(s): 1

31 Which one of the given options is the postfix expression for the prefix expression given below?

32 ★ $*+AB-CD$ 33 ☐ $ABCD+-*$ 34 ☐ $AB+-CD*$

35 ☆

36 ☒ $AB+CD-*$ 37 ☐ $AB+CD*-$

38

Previous Question

Next Question

TIME LEFT: 00:08:17

32 ★

QUESTION: 37 GROUP: Programming Basics SECTION: Computer Architecture Mark(s): 1

33

If the number of segments in a pipeline is 6, what is the time taken to execute 75 tasks if the clock cycle time of the pipeline is 15ns?

34

☐ 1099ns

35

☐ 1000ns

36 ★

☒ 1200ns

37 ☆

38

☐ 1500ns

39

40

~~★~~ $(6 + 75 - 1) \times 15$

✓

32 ★ int main(void)

{

33

char a = -3;

34

char b = 253;

if(a == b)

35

{

a = a && -1;

36 ★

printf("%d ", a);

}

37

if(a == 1)

38 ☆

{

b = b >> -1;

39

printf("%d ", b);

}

40

return 0;

}

?

.

32  `b = b >> -1;`

33 `printf("%d ", b);`

34 `return 0;`

35
36  ☐ Compile time error

37 ☐ 0

38  ☐ No output

39 ☒ 1 -1

40
[Previous Question](#)

[Next Question](#)

TIME LEFT: 00:04:08

QUESTION 39 GROUP: Programming Basics SECTION: Computer Architecture Mark(s): 1

A control unit takes input from multiple input sources for generating a signal. Which one of the signal given below is NOT produced as an output?

☐ Memory Module Control Signal

☐ Data Movement Control Signal

☒ Clock Control Signal

☐ I/O Module Control Signal

[Unanswer](#)

[Previous Question](#)

[Next Question](#)

Identify the likely cause of the error in the given code and choose the correct option to fix it.

```
#include <stdio.h>
```

```
int main() {
```

```
    int *ptr; = malloc
```

```
    *ptr = 10;
```

```
    printf("Value at ptr: %d\n", *ptr);
```

```
    return 0;
```

```
}
```

☐ No error

TIME LEFT: 00:29:51

41 ☆ QUESTION: 41 GROUP: Computer Science SECTION: Operating Systems Mark(s): 1

42 In which of the Multi threading model given in options the system does not block if a particular thread is blocked?

43 ☐ One to many model

44 ☐ Many to one model

45 ☒ Many to many model ✓

46 ☐ One to one model

47

48

49

Previous Question

Next Question

Choose the correct statement(s) from the given options.

- I) Static methods can only access the static members of the class, and can only be called by other static methods. ✓
- II) Static class's object cannot be created, and it only contains static members. ✓

☐ Only (II)

☐ Only (I)

☐ Neither (I) nor (II)

☒ Both (I) and (II) ✓

Previous Question

Next Question

41 int main(void)

42 {

const unsigned char a = 255;

const signed char b = 255;

int c;

c = fun(a, b);

printf("\n%d %d %d", a, b, c);

return 0;

46 }

int fun(char a, char b)

47 {

printf("%d ", a << 1);

printf("%d ", b << 1);

49 }

41 In the "Open Closed Principle" any new functionality should be implemented by:

- 42 I) adding new classes,
43 II) adding attributes and methods, instead of changing the current ones or existing ones.
44 III) modifying the existing class.

44 ☆

45 ☐ Only (I) and (II)

46 ☐ Only (II)

47 ☐ Only (I)

48 ☐ Only (II) and (III)

49

Previous Question

Next Question

42

```
int i = 10, j = 15;
```

43

```
void Sample(int **q, int **p)
```

```
{
```

44 ★

```
int l = **q;
```

```
*q = *p;
```

45

```
**p = l;
```

46 ★

```
**p *= **q;
```

```
**q += **p;
```

```
};
```

47 ☆

```
int main()
```

```
{
```

48

```
int *p = &i, *q = &j;
```

49

```
Sample(&p, &q);
```

50

```
printf("%d %d", i, j);  
return 0;
```


42

```
int *p = &i, *q = &j;
```

43

```
Sample(&p, &q);
```

```
printf("%d %d", i, j);
```

44 ★

```
return 0;
```

```
}
```

45



0 10

46 ★

programming Mark(s): 1
What is the meaning of the statement given below?

Char *ipointer = NULL;

- ☐ It means a invalid declaration.
- ☐ ipointer is pointing to a garbage value.
- ☐ ipointer is pointing to nothing.
- ☐ ipointer is pointing to the value "zero".

Which of the line(s) generates error in the program given below?

46 ★
#include <stdio.h>

47
int main()

{

48
int apple[2][3] = {2, 4, 6, 8, 10, 12};

49 ★
int *ap = (int *)apple;

//Line 1

int **app = ≈

50 ☆
int ***appl = &app;

//Line 2

++>(*app)++;

51
++++>(*appl);

//Line 3

52
printf("%d", apple[0][1]);

return 0;

53
}

46 ★

Which of the below given statement(s) is/are correct about context switching?

47

48

49 ★

50

I) The context of a process state is represented in the PCB of a process.

II) The system stores and restores the CPU's state such that multiple processes can share a single CPU.

III) Context switching is just limited to thread switching.

☐ Only (I) and (II)

45
46
47
48
49
50
51
52
53

```
++(*app)++;  
++*+{*(app)};  
printf("%d", apple[0][1]);  
return 0;
```

//Line 3

- ☐ No error
 - ☐ Line 1 and Line 3
 - ☐ Line 2
 - ☐ Line 2 and Line 3
- [Previous Question](#) [Next Question](#)

TIME LEFT: 00:13:45

47

QUESTION: 52 GROUP: Computer Science SECTION: Data Structures Mark(s): 1

48

Which set of operations given in options when performed on empty stack 'K' of size 2 will make $TOP = N - 1$ where 'N' is the size of the stack (In case of underflow and overflow conditions further operations are not considered)?

49 ★

☐ POP(), PUSH(K, A), POP(), PUSH(K, T), PUSH(K, E)

50 ★

☐ PUSH(K, G), POP(), POP(), PUSH(K, F)

51

52 ☆

☐ PUSH(K, M), PUSH(), PUSH(K, Y), PUSH(K, U)

53

☐ PUSH(K, H), PUSH(K, W), POP(), PUSH(K, ...)

54

48

Which of the given statement(s) is /are true about the radix integer sort?

49 ★

I) Radix integer sort's efficiency is $O(k \cdot n)$ for n keys which have k or fewer digits.

50 ★

II) The data structure used in Radix integer sort is only queue.

III) There are three classifications of radix sorts (LSD, MSD, NSD radix sorts).

51

49 ★ #include <stdio.h>

int main()

50 ★

{

unsigned char i = 0x80 ;

51

printf("%d", i << 1);

52

return 0;

}

What is the memory location of item $ARR[11][23]$ when the array is stored in row major and column major respectively?

Given array $ARR[35][45]$ requires 4 byte for every element and base memory location is 3280.

Note row-major order storage, a multidimensional array in linear memory is organized such that rows are stored one after the other.
column-major order is a similar method of flattening arrays onto linear memory, but the columns are listed in sequence.]

☐ 6244 and 7152

☐ 5352 and 6544

☐ 4292 and 4292

☐ 4912 and 7464

Previous Question

Next Question

52

QUESTION: 59 GROUP: **Computer Science** SECTION: **OOPS** Mark(s): 1

53 ★

Choose the incorrect statements from the given options.

54 ★

I) Static variables live as long as their object lives.

II) Instance variables live basically as long as their class lives.

III) Local variables live as long as their method is on the stack; however, if their method invokes another method, they are temporarily unavailable.

IV) Block variables live until the block completes.

☐ Only (I) and (IV)

```
char *a = (char *)0x00000120;
```

```
char *b = (char *)0x00100020;
```

```
char *c = (char *)0x00120000;
```

```
if(a == b)
```

```
    printf("\nHello");
```

```
if(a == c)
```

```
    printf("\nHi");
```

```
if(a > b && a > c && b > c)
```

```
    printf("\nBye");
```

```
else
```

```
    printf("\nDefault");
```

```
return 0;
```

Hi

```
printf("\nDefault");
```

```
return 0;
```

☐ Hi

☐ Hello

☐ Bye

☐ Default

[Previous Question](#)

[Next Question](#)